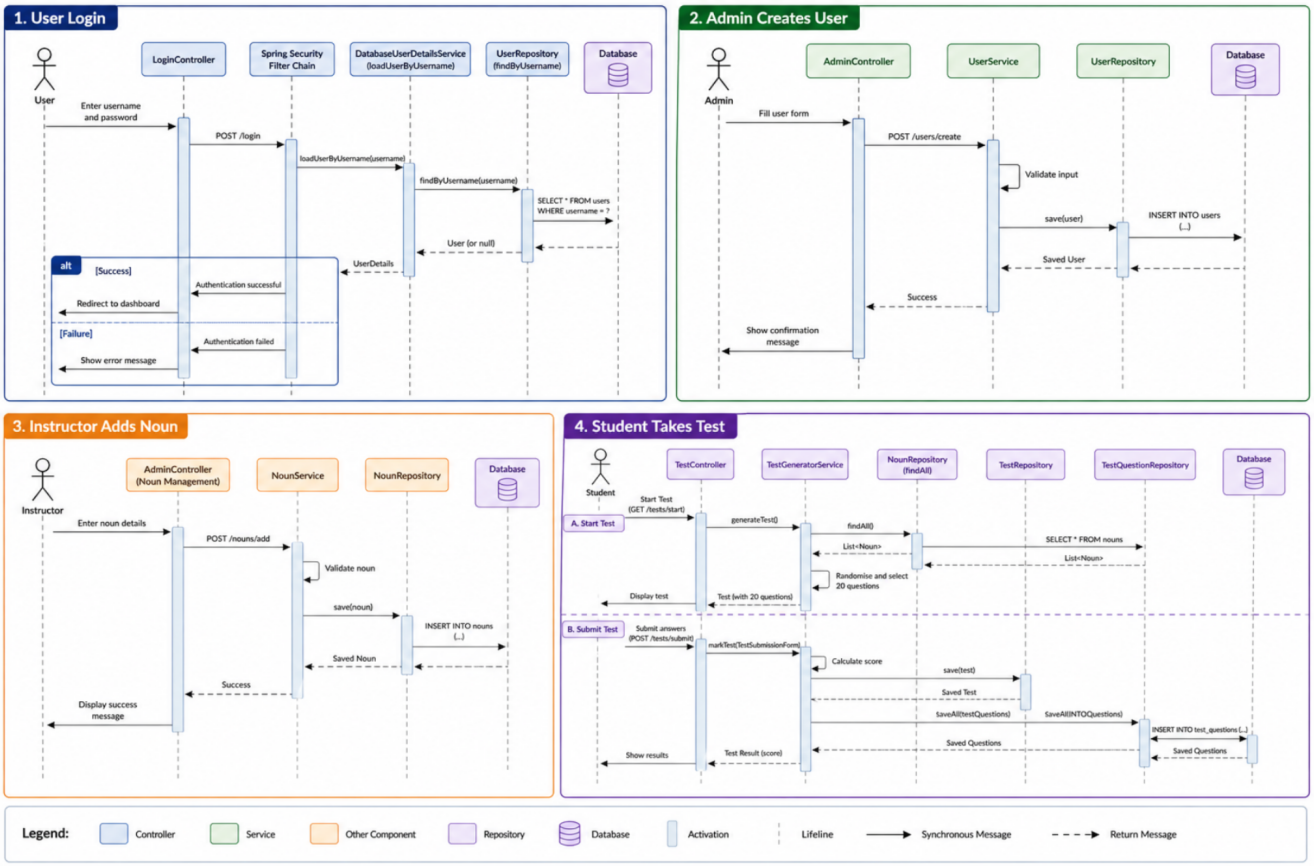
1. User Login
User
LoginController
Spring Security Filter Chain
DatabaseUserDetailsService (loadUserByUsername)
UserRepository (findByUsername)
Database
Enter username and password
POST /login
loadUserByUsername(username)
findByUsername(username)
SELECT * FROM users WHERE username = ?
User (or null)
UserDetails
alt
[Success]
Authentication successful
Redirect to dashboard
[Failure]
Authentication failed
Show error message
2. Admin Creates User
Admin
AdminController
UserService
UserRepository
Database
Fill user form
POST /users/create
Validate input
save(user)
INSERT INTO users (...)
Saved User
Success
Show confirmation message
3. Instructor Adds Noun
Instructor
AdminController (Noun Management)
NounService
NounRepository
Database
Enter noun details
POST /nouns/add
Validate noun
save(noun)
INSERT INTO nouns (...)
Saved Noun
Success
Display success message
4. Student Takes Test
Student
TestController
TestGeneratorService
NounRepository (findAll)
TestRepository
TestQuestionRepository
Database
A. Start Test
Start Test (GET /tests/start)
generateTest()
findAll()
SELECT * FROM nouns
List<Noun>
List<Noun>
Randomise and select 20 questions
Test (with 20 questions)
Display test
B. Submit Test
Submit answers (POST /tests/submit)
markTest(TestSubmissionForm)
Calculate score
save(test)
Saved Test
saveAll(testQuestions)
saveAll(INTOQuestions)
INSERT INTO test_questions (...)
Saved Questions
Saved Questions
Test Result (score)
Show results
Legend:
Controller
Service
Other Component
Repository
Database
Activation
Lifeline
Synchronous Message
Return Message